

PAYAL JAIN

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PROFESSIONAL EXPERIENCE

Accenture Solutions Pvt Ltd.: AI/ML computational Science Sr Analyst: Feb 2024 – present

- Led the development of a GenAI based Presentation Creator Tool:
 - Designed and implemented a scalable GenAI solution using OpenAI GPT models, enabling dynamic generation of business presentations based on topic, template, and contextual document input.
 - Utilized langchain to load, chunk, and retrieve data from user-uploaded files and client-specific knowledge bases, supporting Retrieval-Augmented Generation (RAG) workflows.
 - Applied advanced prompt engineering to ensure consistent and structured outputs for template recommendations, agenda suggestions, and content generation tailored to business contexts.
 - Integrated Milvus vector database (hosted via Docker on Azure VM) for chunk-based document storage and semantic search, enhancing the system's retrieval capabilities.
 - Built and deployed production-grade components on Azure using services such as Azure OpenAI, Azure VM, and CI/CD pipelines, ensuring reliability and maintainability.
 - Collaborated cross-functionally with frontend and PR teams and led architectural design to align with business goals and user experience expectations.
- Intelligent Application Portfolio Management:
 - Enhanced the existing LOB recommendation engine, which initially relied on fuzzy matching and keyword analysis, by integrating a GenAI-based model to improve accuracy.
 - Contributed to the feature utilizing Sentence Transformer with BERT for generating embeddings of application descriptions, utilizing cosine similarity scoring to accurately map them to the appropriate business capabilities.
 - Utilized Azure Blob Storage to manage and access large-scale database files and employed the Pandas library for merging and transforming datasets during preprocessing.
- Retrieval-Augmented Generation (RAG)-based chatbot.
 - Developed advanced features such as chat history management, reply to previous queries option for users, ensuring seamless conversational flow and enhance user interaction.
 - Integrated functionality to suggest new queries to users.
- Exploration of Multimodal and Agentic AI Capabilities
 - Conducted hands-on experimentation with state-of-the-art multimodal LLMs including Gemini, llama, and GPT-4, evaluating their capabilities for text, image and video inputs.
 - Completed training on Accenture's Agentic AI Framework, gaining practical knowledge in designing autonomous LLM agents capable of tool use, task orchestration

Ineda Systems Pvt. Ltd.: Senior Design Engineer (SDE): April 2016 – April 2018

Automated driving using Machine Learning

- Radar based target classification:
 - Led the design and implementation of High-performance predictive ML model for target classification.
 - Demonstrated proficiency in extracting analytical insights of data collected from FMCW radar using **Matplotlib and Seaborn libraries**.
 - Implemented techniques to extract valuable statistical features from the interpreted data.
- Implemented several ML classification techniques like SVM (using all kinds of kernels), Random Forest, XGBoost, Principal Component Analysis using Numpy, Scikit-learn.

Embedded Systems

- Self-training on embedded components of the SoCs delivered by Ineda Systems.
- Developed 8bit OLED display for a SoC at app, interface, engine, display driver level.
- Fixed bugs and implemented test cases for MPU and MCU components of the SOC's.

Samsung India Electronics Pvt Ltd – NOIDA: Software Engineer: July 2012 - Dec 2015

- Worked on coding and development of Application of feature phones:
 - Developed new GUI screens and functionalities for FM Radio features of new phone models.
 - **Proposed new ideas and enhanced FM radio app** by adding RDS support, Alternate Frequency, background FM recording, FM sharing via Bluetooth and SMS.
- Worked in White box team with the major responsibilities of code review, test-case development, code optimization, prevention of new issues and side-effects due to code modifications for Contacts and FM.
 - **Size Optimization of .ini files storing** from 31MB to 64KB for the basic feature phones.

Professional Development (during career break from April 2018- Feb 2024)

- Completed "NLP Specialization" by deeplearning.ai on Coursera.
- Completed certification course - **Deep Learning: Advanced Natural Language Processing and RNNs** on Udemy.

SKILLS

- ✓ **Python** (NumPy, Pandas, Scikit-Learn, Tensorflow, Kears, Matplotlib, NLP, NLTK)
- ✓ Machine Learning, Deep Learning
- ✓ Azure Devops, Azure services (VM, Blob storage, Azure OpenAI, Function App)
- ✓ Flask API, Fast API
- ✓ Gen AI (LLM models – OpenAI(3.5 turbo, 4.o, embedding model), Gemini, llama 3.2, Claude), RAG, vector database (Milvus),
- ✓ Programming in C/C++, Embedded Systems
- ✓ SQL Basics
- ✓ ANN, CNN, RNN, LSTM, GRU

PROFICIENCY

Software Tools: VS Code, Anaconda, Review Board, Git, Google Colab, Postman

Operating system: Windows, Ubuntu

ACADEMICS

B.Tech: Computer Science and Engineering (2010-2012) **CGPA** - 4.82/5(85.1%) NERIST University,Itanagar

Diploma: Computer Science and Engineering (2008-2010) **CGPA** - 4.62/5(81.9%) NERIST University,Itanagar

ACHIEVEMENTS & AWARDS

- ✓ Awarded **Gold Medal** in BTech (Computer Science and Engineering) in North Eastern Regional Institute of Science and Technology.
- ✓ Received multiple Awards for early deliverables and performance excellence at Samsung Research Institute, Noida